

United Kingdom

Competition and Markets Authority

I. Analytics and machine learning in evidence review

Throughout 2023, the CMA has continued to roll out its new evidence review platform. eDiscovery specialists and data scientists from the Data Technology and Analytics ('DaTA') unit have worked together to help the CMA's case teams exploit new analytical techniques to speed up evidence review. These include Continuous Active (Machine) Learning, email threading and clustering of similar documents. We continue to look for innovative ways to exploit our evidence review technology and, most recently, we have piloted the use of our evidence review platform to agree confidentiality representations with external parties on a cartel investigation.

The CMA is increasingly receiving submissions from parties, for merger and other cases, with documents that are not in English. While we do use professional translators, given the scale and its impact on both cost and time, we needed a solution that would allow us to quickly process thousands of documents and allow case teams to run keyword searches and review the documents as quickly as possible. Given the confidential nature of these business documents, using an external automated translation service is not appropriate. The DaTA team therefore took an open-source solution using NLP and adapted it to run on our secure data analytics platform, adding parallelism, which allowed us to auto-translate documents at scale.

II. Open-sourcing our web-scraping

Mirroring the move by business to digital, the DaTA team often finds itself scraping information from the web. The uses of scraping range from preliminary evidence collection all the way to automating remedy compliance monitoring. The CMA implements web-scraping in a careful and considerate manner, ensuring that it is proportionate, legal and does not over burden the company's servers. The DaTA unit has put together a Python package, called "selene", that standardizes our internal scraping code, by providing an (object-oriented) framework, based on open-source web-scraping packages. This approach reduces the amount of duplicated code for standard tasks from scrolling to taking screenshots. The DaTA unit has open sourced this package and publicized it widely with the hope that it can both be useful to other agencies, but also be the start of a more open and collaborative code-driven future to regulation.

III. Automating the identification of mergers and acquisitions

In the UK there is no requirement for parties to a merger to notify the CMA of a transaction. The CMA, therefore, needs to monitor news sources to identify mergers and assess whether they raise competition concerns. This has historically been a largely manual process at the CMA. The DaTA unit have recently developed a tool to track merger activity in an automated way with machine learning. The tool, which is still in development, collects news article data from various APIs and RSS feeds and passes them through a natural language machine-learning model. It then outputs a prediction as to whether each article relates to a merger. The tool should allow the CMA to dramatically reduce the amount of time it takes to sift through long lists of news articles for any sign of a relevant merger, freeing up the team to focus more on analysis of the competition concerns of each merger identified.

IV. Building our behavioral science capability

Throughout 2022, we have continued to build our Behavioral Hub within the DaTA unit. In 2023, we published a paper on experiments at the CMA which describes how and when we use field and online experiments to support our work, including in relation to online choice architecture.⁴⁵ We continue to explore and identify harm from online choice architecture, including across our portfolio of consumer protection and antitrust cases and markets work. To support this, our behavioral and data scientists have been working together to set up an eye-tracking lab to test how users interact with websites.

V. Horizon-scanning for emerging technologies

The CMA has taken steps to ensure it is proactive and forward-looking in the emergence of new technologies or emerging markets. This includes the development of its internal horizon-scanning capabilities to identify new and emerging technologies and trends in digital markets. For shortlisted technologies, our DaTA unit have produced ‘technology primers’ that explain the technology, the markets to which it is most relevant and the possible implications for competition and consumer protection.

⁴⁵ GOV.UK, *Experiments at the CMA: How and when the CMA uses field and online experiments*, <https://www.gov.uk/government/publications/experiments-at-the-cma-how-and-when-the-cma-uses-field-and-online-experiments> (last accessed June 7, 2023).

Large language models, or generative AI, was identified through our horizon-scanning in 2022. In May 2023, following a period of internal research, the CMA opened a review on large artificial intelligence (AI) models to understand how the rapidly evolving sector could help people, businesses, and the UK economy. The review will help the CMA develop its understanding of large AI models, known as foundation models, and their future uses in a wide range of markets as well as any competition or consumer issues that might arise.